

8. Write the python program to implement DFS.

Aim

To develop a Python program to implement the Depth-First Search (DFS) algorithm for traversing a graph starting from a user-specified source node.

Algorithm

1. Start the program.
 2. Read the number of vertices and edges from the user.
 3. Create an adjacency list to represent the graph.
 4. Read all the edges and store them in the adjacency list.
 5. Read the starting vertex.
 6. Initialize an empty visited set.
 7. Visit the starting vertex and mark it as visited.
 8. Display the current vertex.
 9. Recursively visit each unvisited adjacent vertex.
 10. Continue until all reachable vertices are visited.
 11. Stop.
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Program

```
# Function to perform DFS
def dfs(graph, vertex, visited):
    visited.add(vertex)
    print(vertex, end=" ")

    for neighbor in graph[vertex]:
        if neighbor not in visited:
            dfs(graph, neighbor, visited)

# Main Program
n = int(input("Enter the number of vertices: "))
e = int(input("Enter the number of edges: "))
```

```
# Create adjacency list
graph = {}

for i in range(n):
    vertex = input(f"Enter vertex {i+1}: ")
    graph[vertex] = []

# Read edges
print("\nEnter the edges:")
for i in range(e):
    u, v = input().split()
    graph[u].append(v)
    graph[v].append(u)      # Remove this line for directed graph

# Read starting vertex
start = input("\nEnter the starting vertex: ")

visited = set()

print("\nDFS Traversal:")
dfs(graph, start, visited)
```

Input

Enter the number of vertices: 5

Enter the number of edges: 5

Enter vertex 1: A

Enter vertex 2: B

Enter vertex 3: C

Enter vertex 4: D

Enter vertex 5: E

Enter the edges:

A B

A C

B D

C E

D E

Enter the starting vertex: A

Output

```
DFS Traversal:  
>>> D E C
```

Note: The DFS traversal order may vary depending on the order in which the edges are entered.

Result

The Python program was successfully implemented to perform Depth-First Search (DFS) on a graph. The program accepts the graph and the starting vertex from the user, traverses the graph using the recursive Depth-First Search algorithm, and displays the DFS traversal order.